

Student Mental Health Monitoring System

SE LAB Experiment-04B | BATCH-D

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Experiment Objectives

4. Realize use cases using **Sequence Diagrams**—system functionality distributed to analysis classes.
5. Realize use cases using **Activity Diagrams**—all normal, alternative, and exceptional flows.
6. Realize use cases with **Control & Class Diagrams**—structural analysis classes.

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UC-01 & UC-02 — Register / Login and Fill Standardised Screening

1. Sequence Diagrams

Sequence Diagram — UC-01: Register / Login

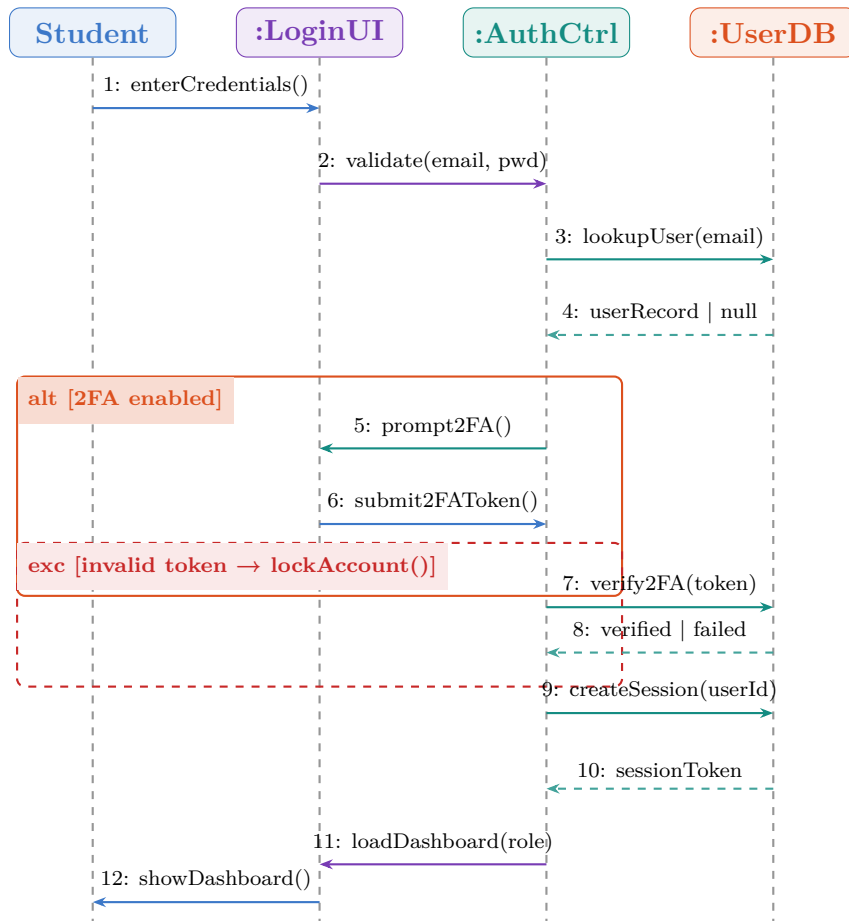


Figure 1: Sequence Diagram — UC-01: Register / Login

Sequence Diagram — UC-02: Fill Standardised Screening (PHQ-9 / GAD-7)

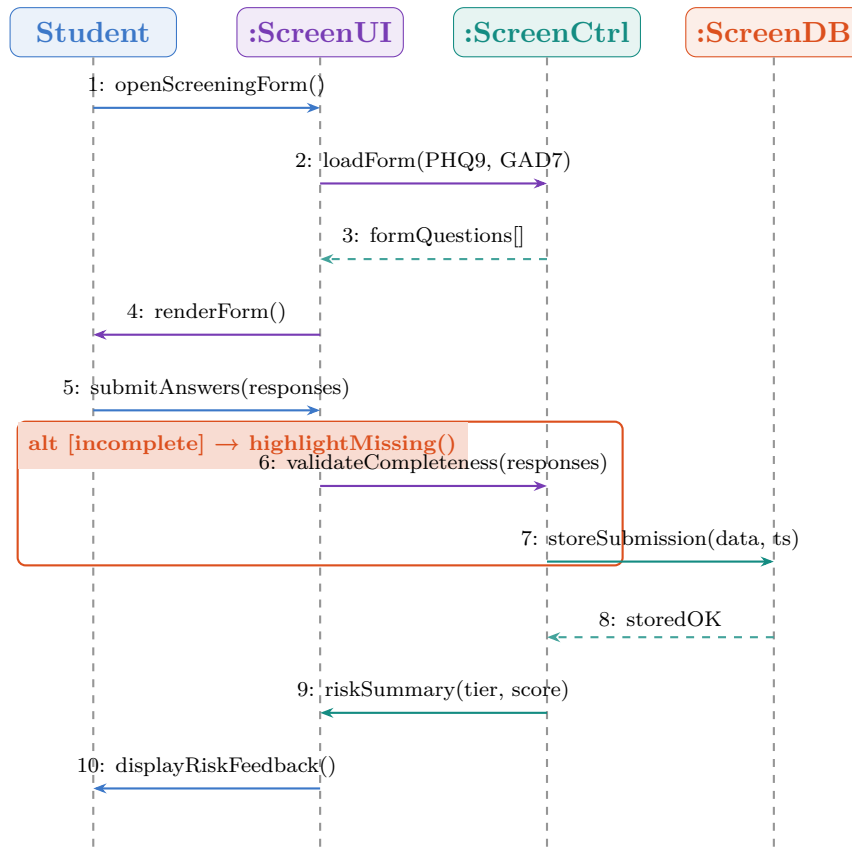


Figure 2: Sequence Diagram — UC-02: Fill Standardised Screening

2. Activity Diagrams

Activity Diagram — UC-01: Register / Login

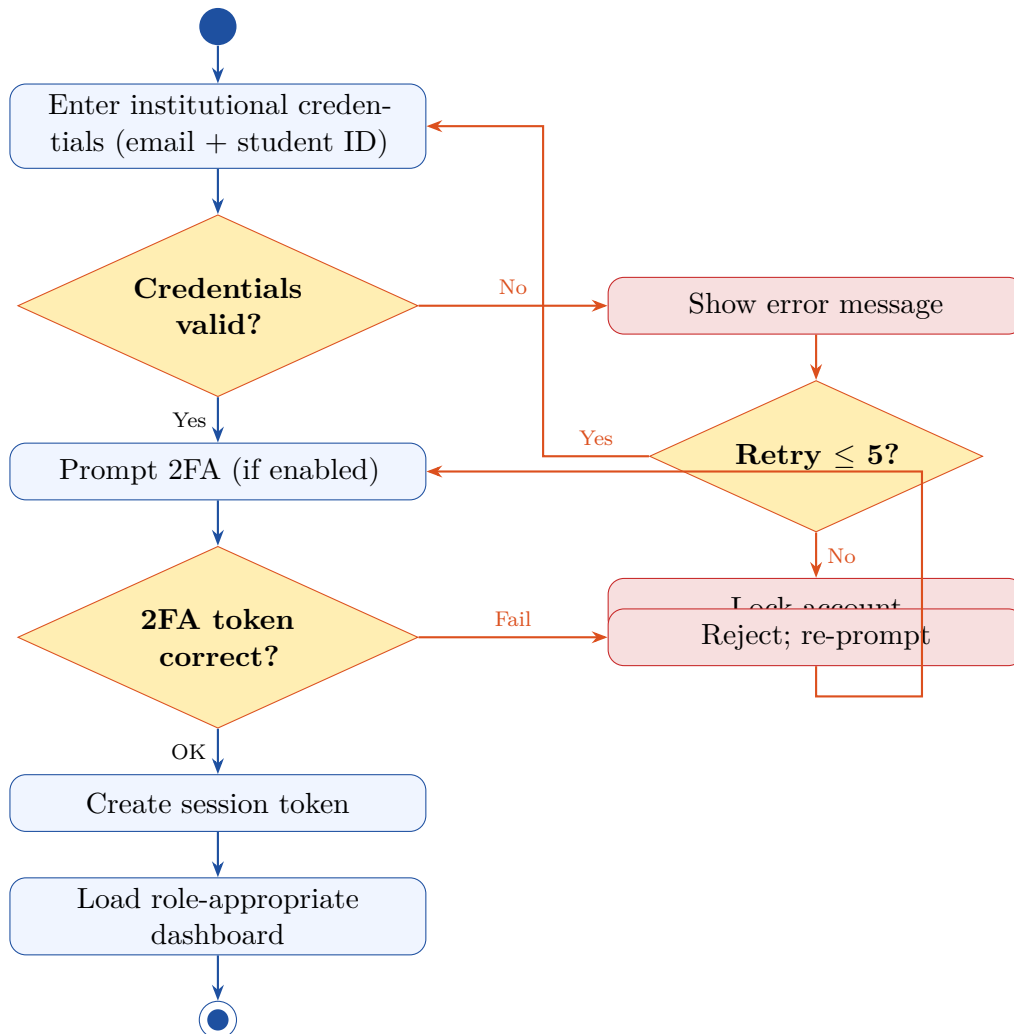


Figure 3: Activity Diagram — UC-01: Register / Login

Activity Diagram — UC-02: Fill Standardised Screening

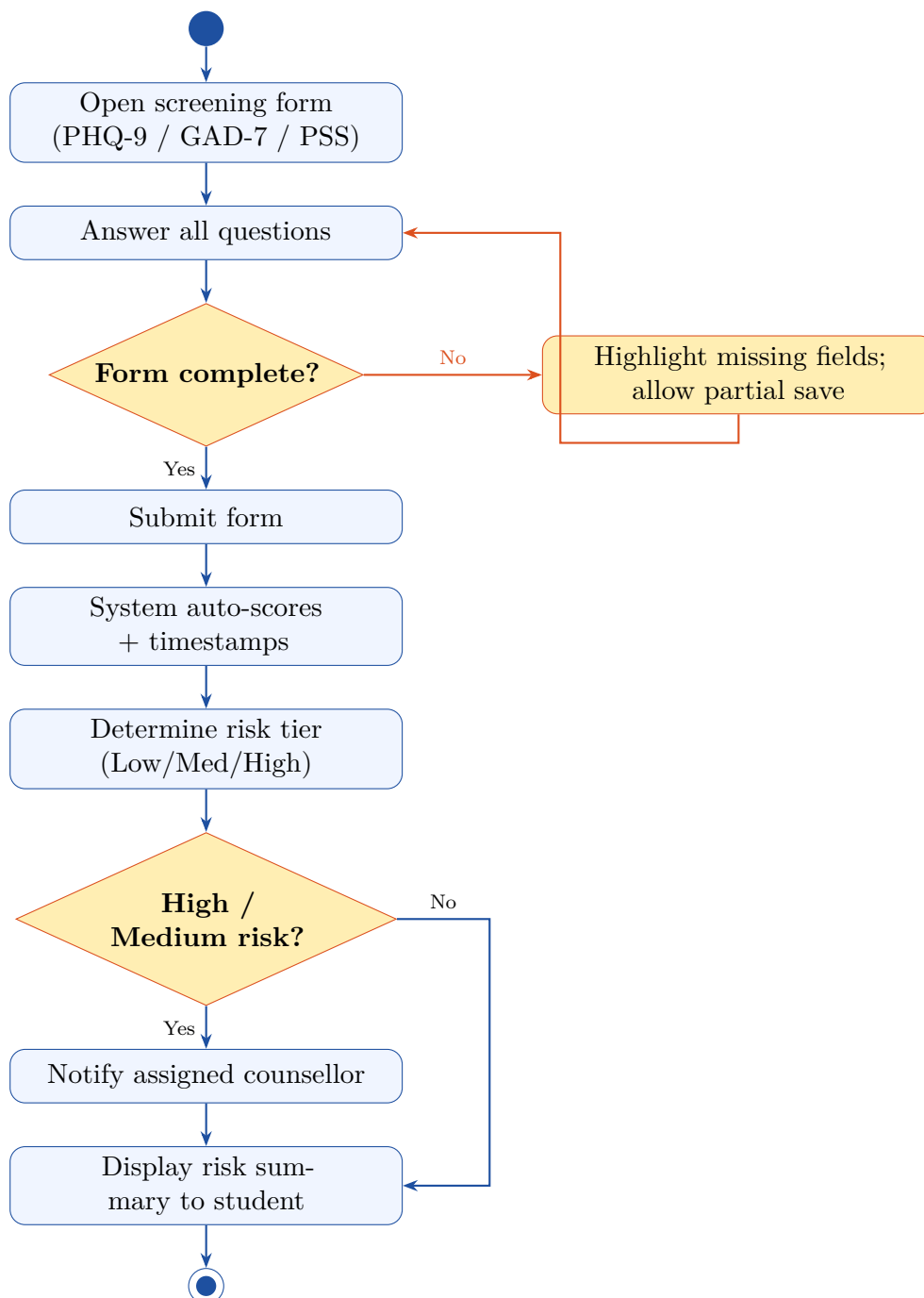


Figure 4: Activity Diagram — UC-02: Fill Standardised Screening

3. Control & Analysis Class Diagram — UC-01 & UC-02

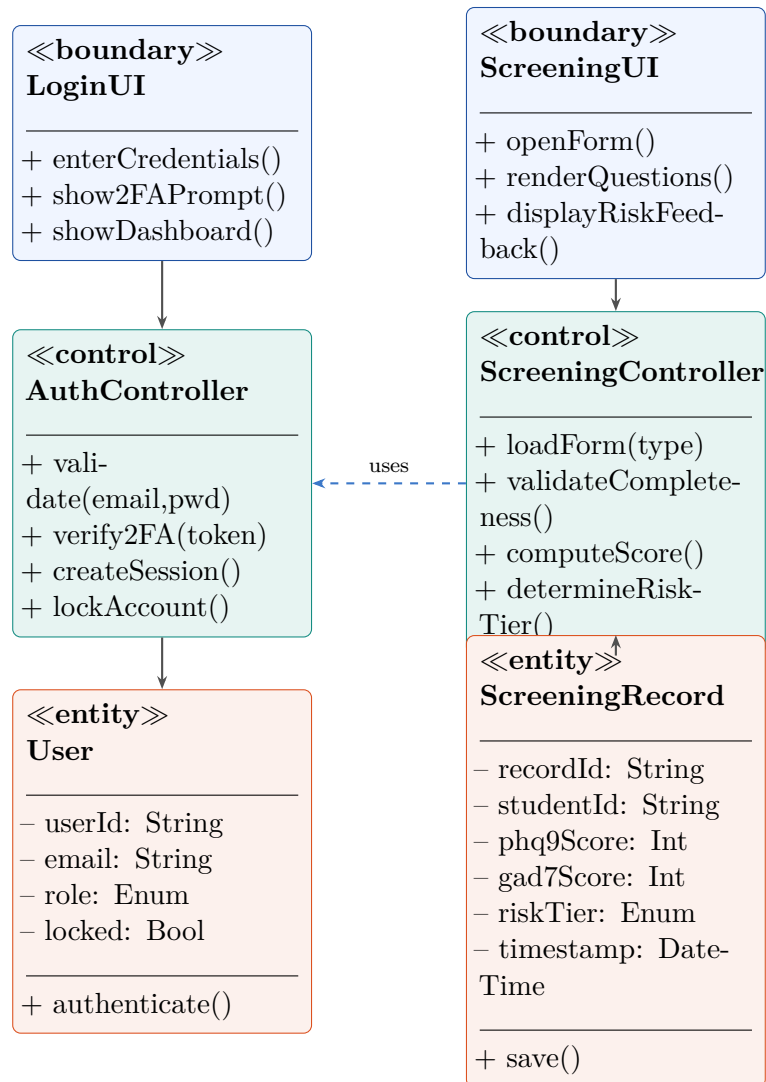


Figure 5: Analysis Class Diagram — UC-01 & UC-02

UC-03 & UC-04 — Log Daily Mood and View Mental Health Status

1. Sequence Diagrams

Sequence Diagram — UC-03: Log Daily Mood & Behaviour

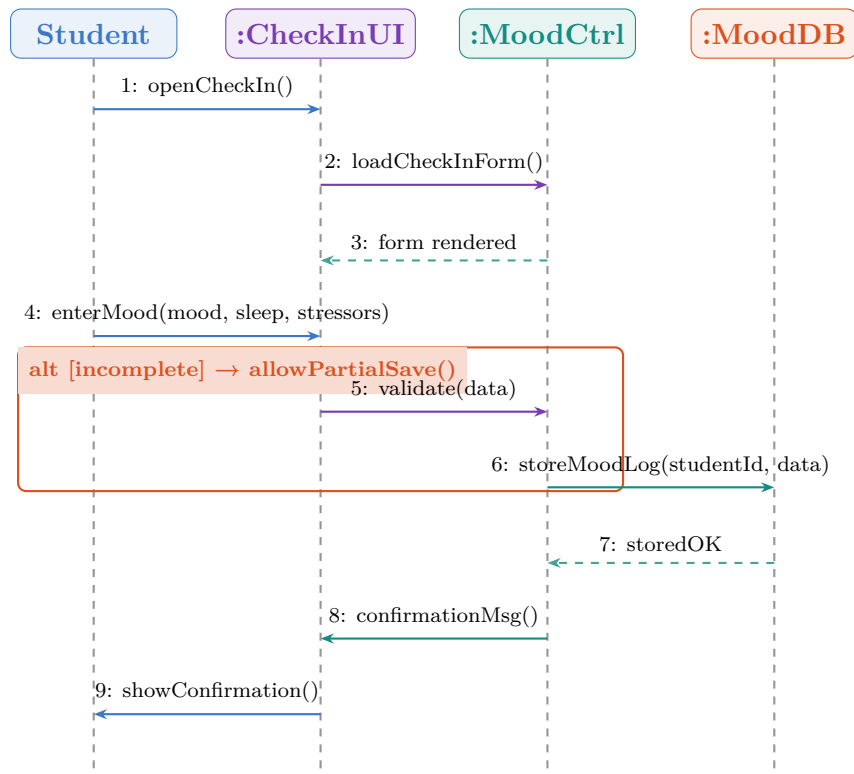


Figure 6: Sequence Diagram — UC-03: Log Daily Mood

Sequence Diagram — UC-04: View Mental Health Status

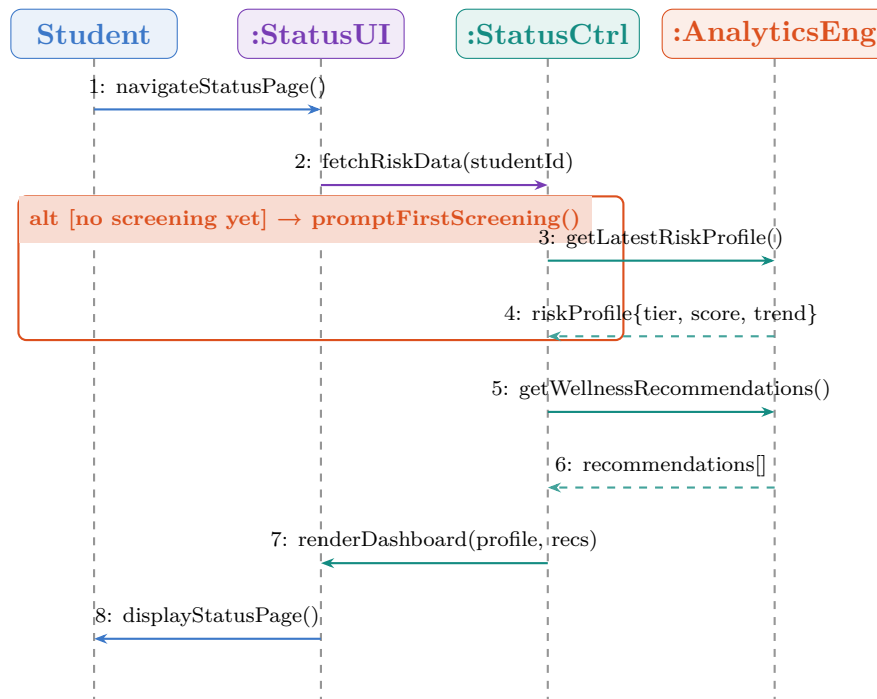


Figure 7: Sequence Diagram — UC-04: View Mental Health Status

2. Activity Diagrams

Activity Diagram — UC-03: Log Daily Mood & Behaviour

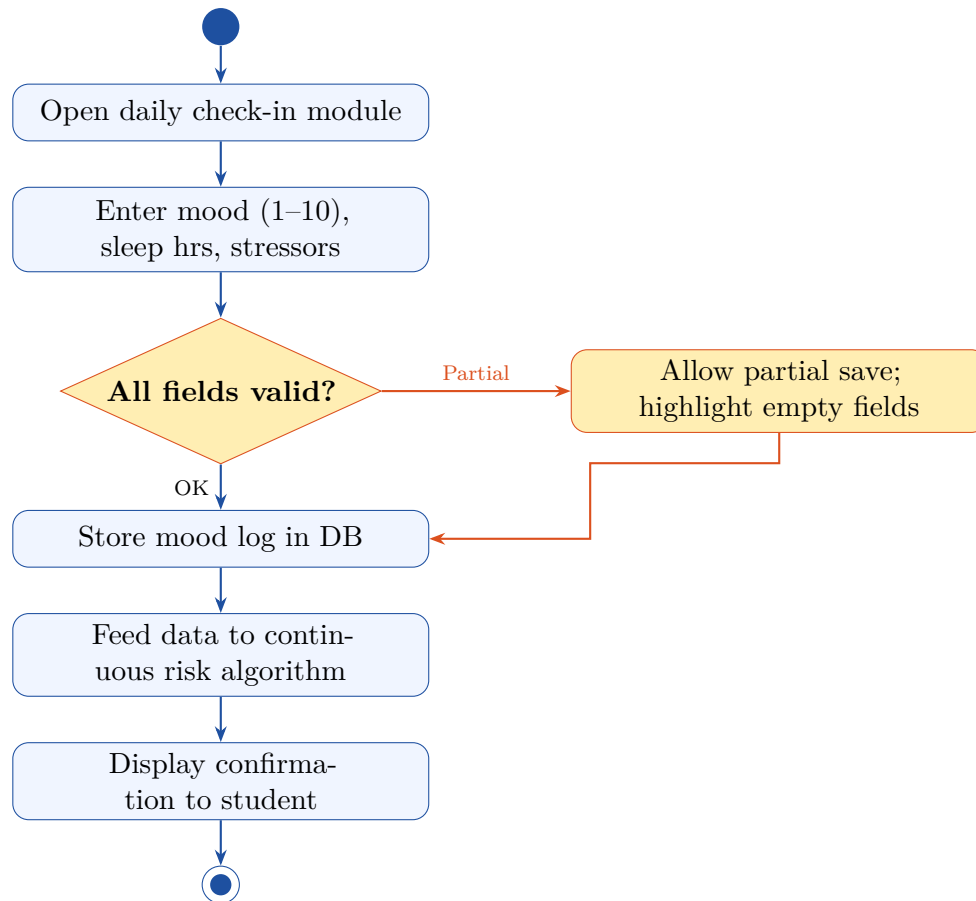


Figure 8: Activity Diagram — UC-03: Log Daily Mood

Activity Diagram — UC-04: View Mental Health Status

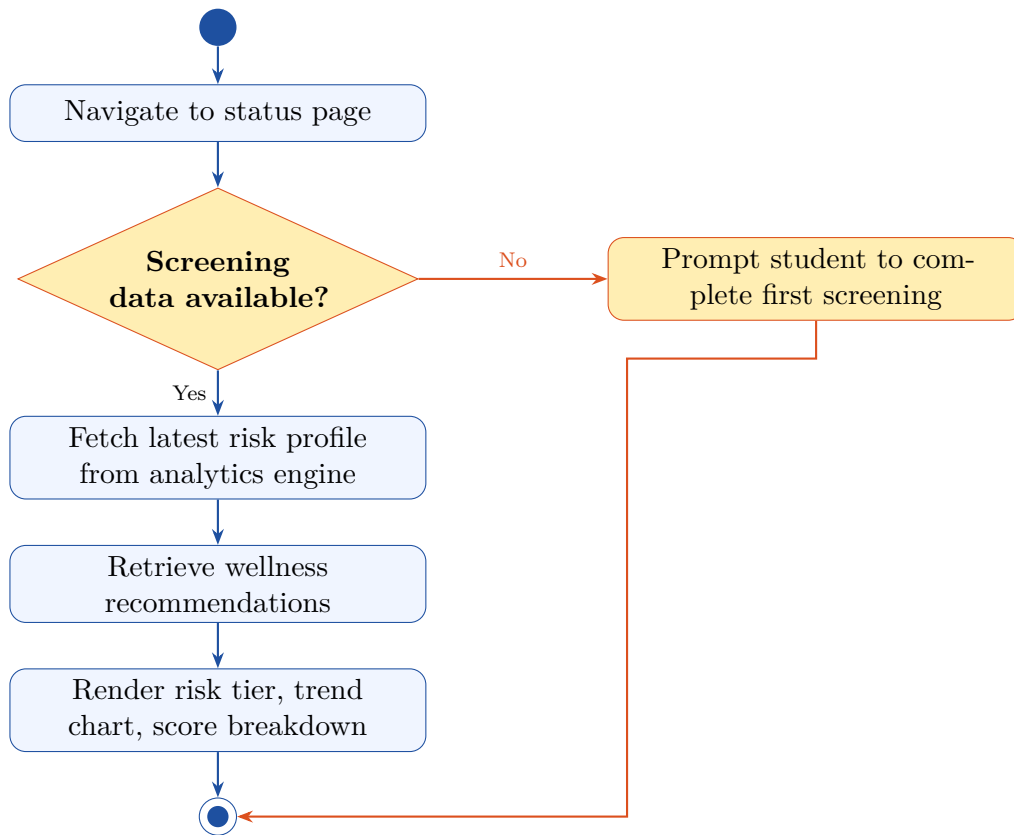


Figure 9: Activity Diagram — UC-04: View Mental Health Status

3. Control & Analysis Class Diagram — UC-03 & UC-04

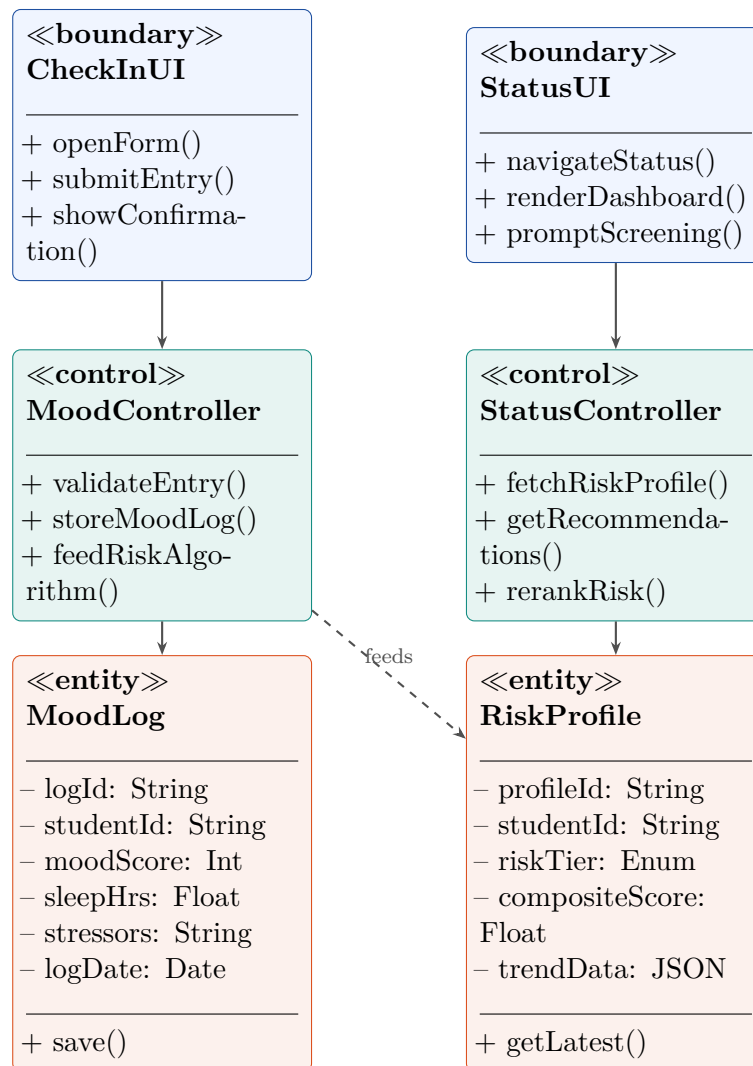


Figure 10: Analysis Class Diagram — UC-03 & UC-04

UC-05 & UC-06 — Request Counselling Session and Receive Notifications

1. Sequence Diagrams

Sequence Diagram — UC-05: Request Counselling Session

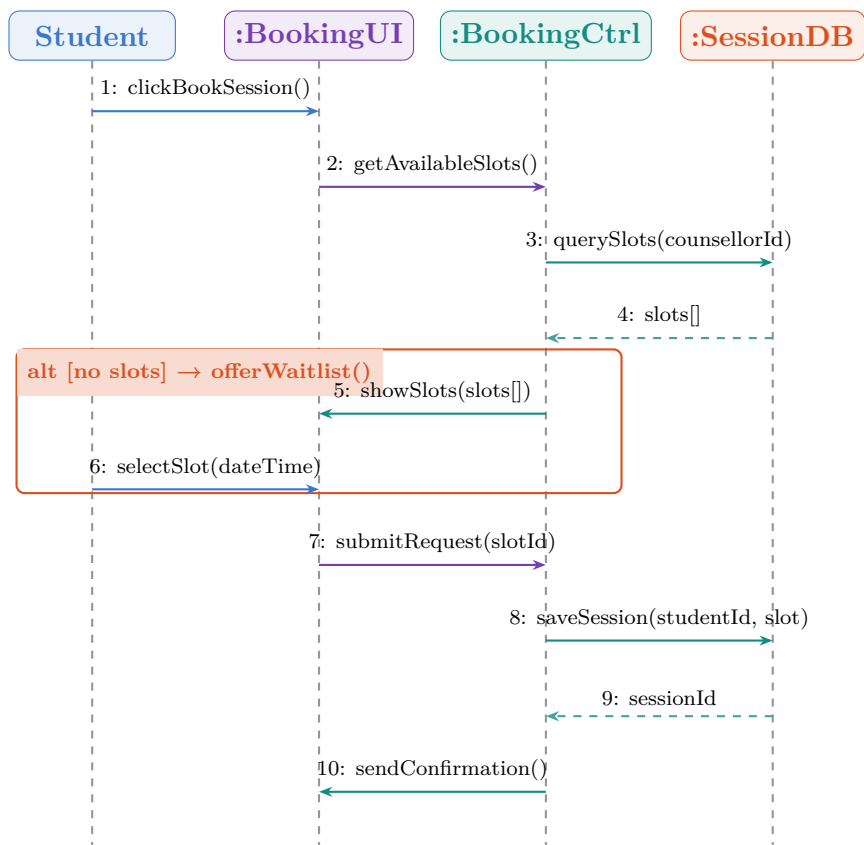


Figure 11: Sequence Diagram — UC-05: Request Counselling Session

Sequence Diagram — UC-06: Receive Notifications & Alerts

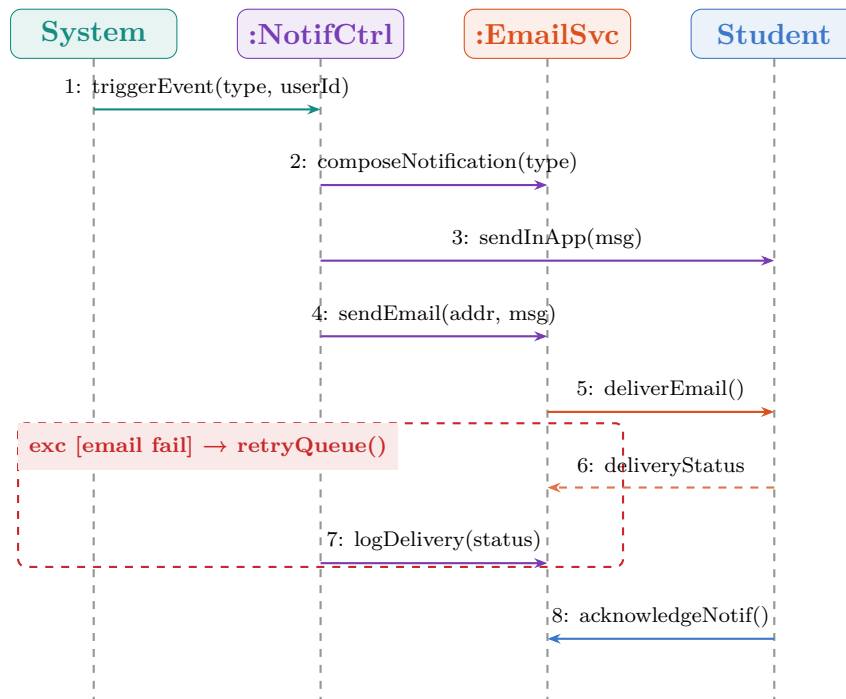


Figure 12: Sequence Diagram — UC-06: Receive Notifications & Alerts

2. Activity Diagrams

Activity Diagram — UC-05: Request Counselling Session

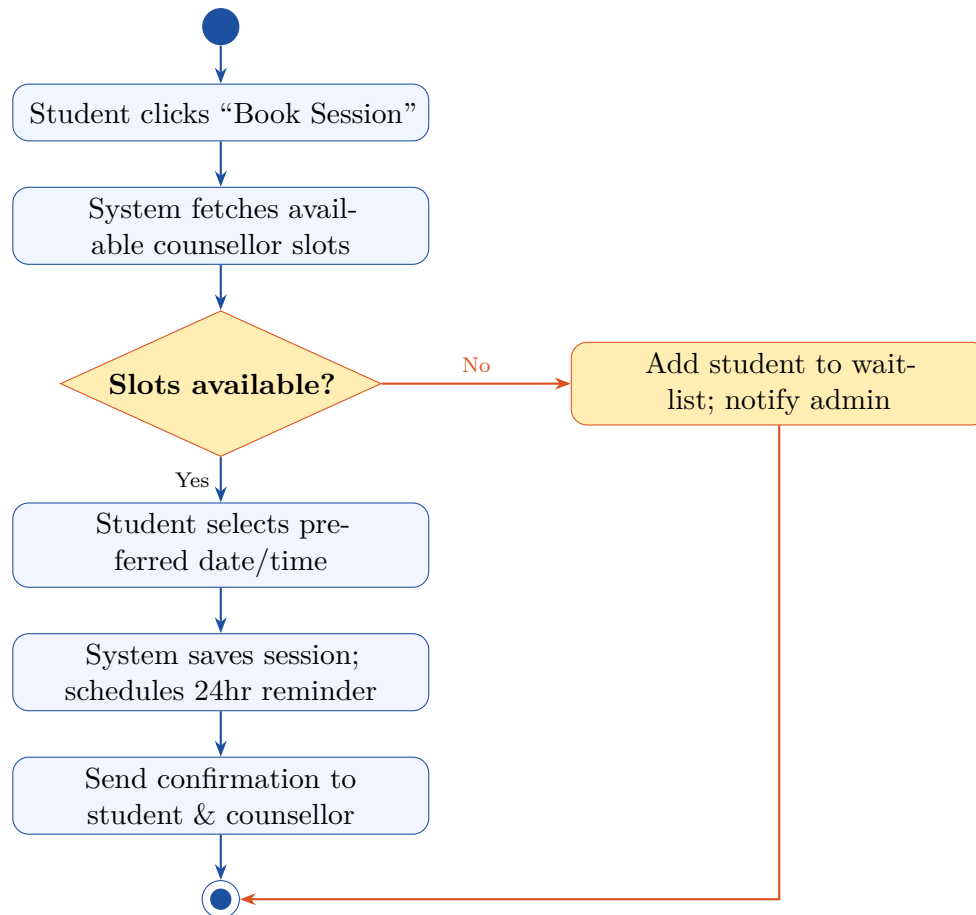


Figure 13: Activity Diagram — UC-05: Request Counselling Session

Activity Diagram — UC-06: Receive Notifications & Alerts

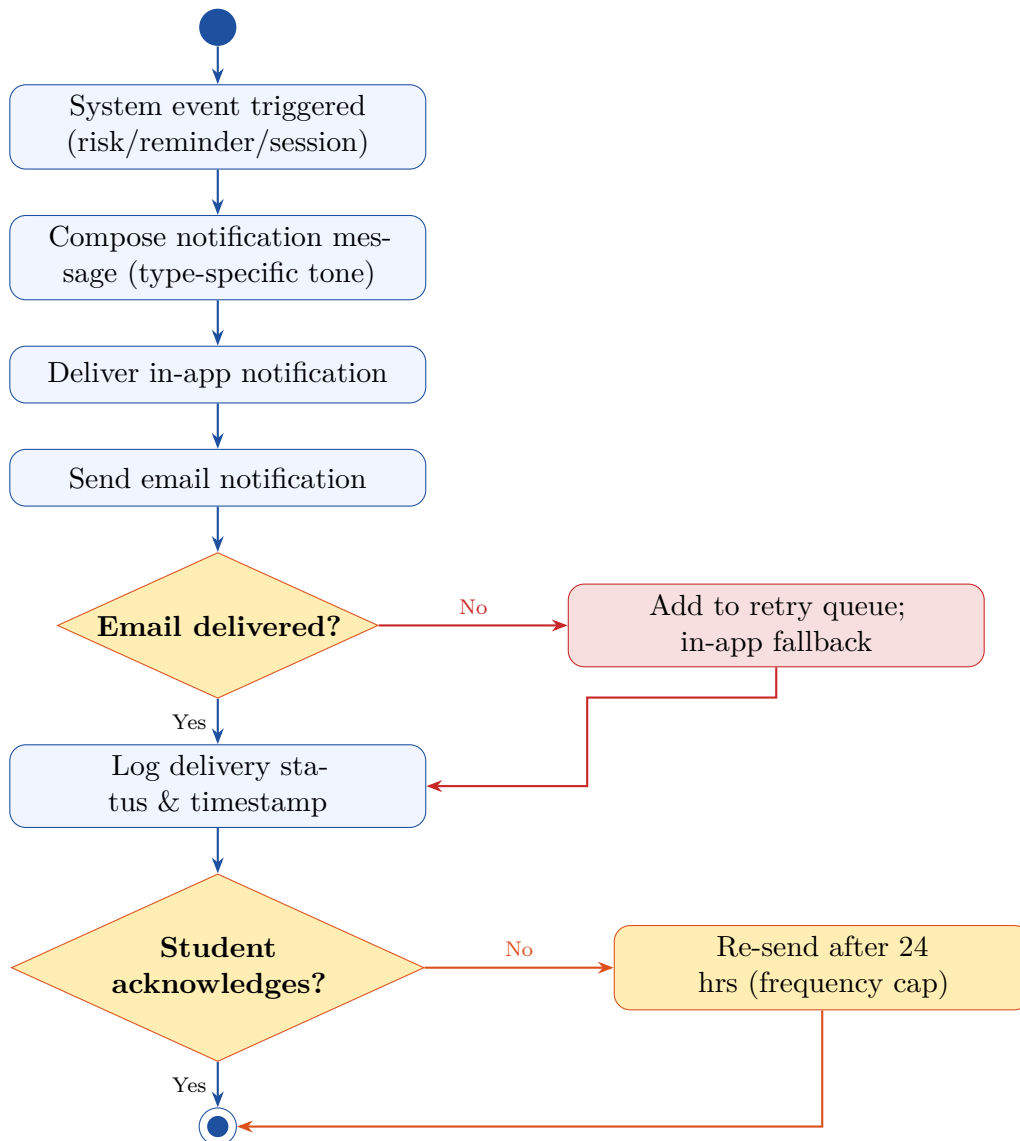


Figure 14: Activity Diagram — UC-06: Receive Notifications & Alerts

3. Control & Analysis Class Diagram — UC-05 & UC-06

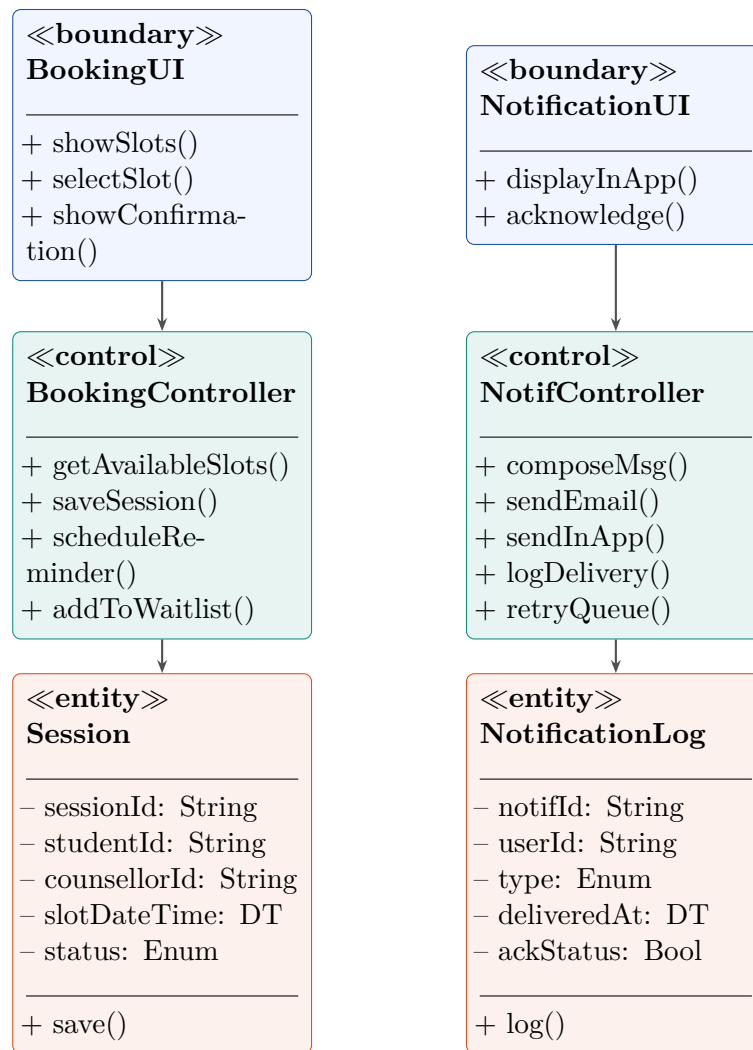


Figure 15: Analysis Class Diagram — UC-05 & UC-06

UC-07 & UC-08 — View Well-being Summary (Parent) and Collect Student Data

1. Sequence Diagrams

Sequence Diagram — UC-07: View Well-being Summary (Parent / Guardian)

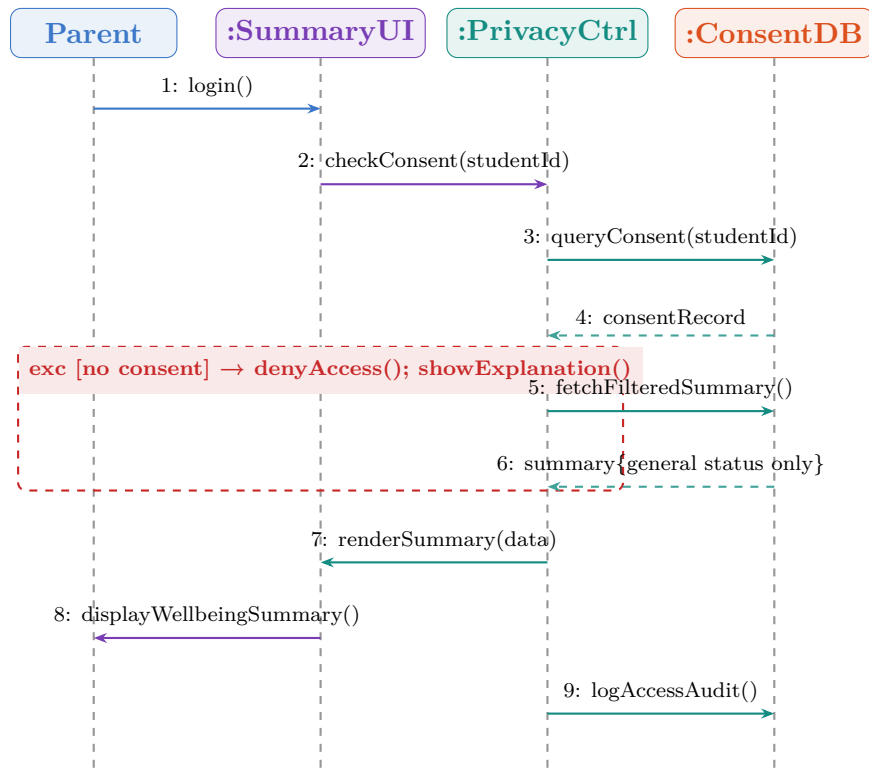


Figure 16: Sequence Diagram — UC-07: View Well-being Summary

Sequence Diagram — UC-08: Collect Student Data

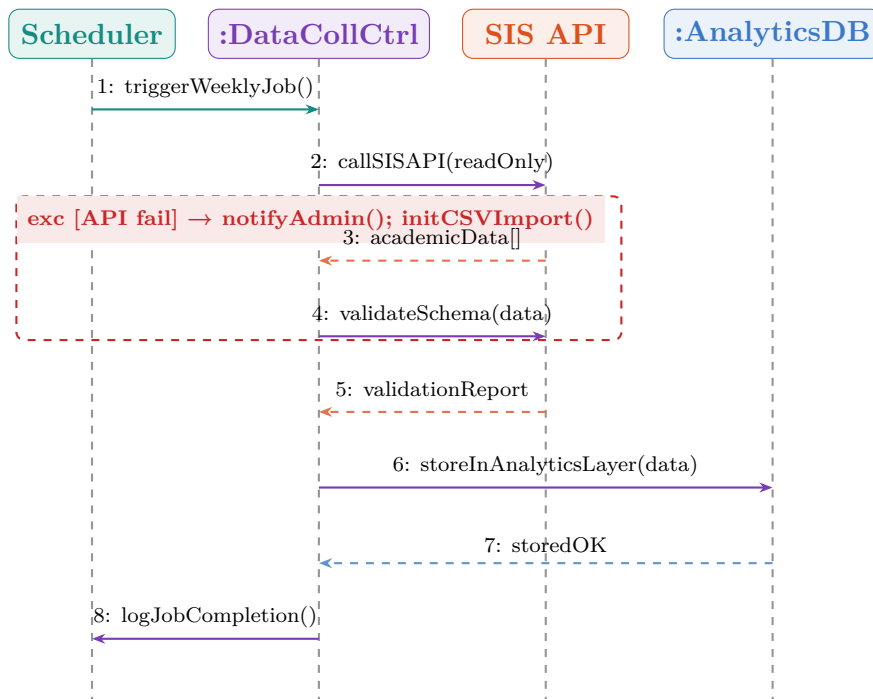


Figure 17: Sequence Diagram — UC-08: Collect Student Data

2. Activity Diagrams

Activity Diagram — UC-07: View Well-being Summary (Parent / Guardian)

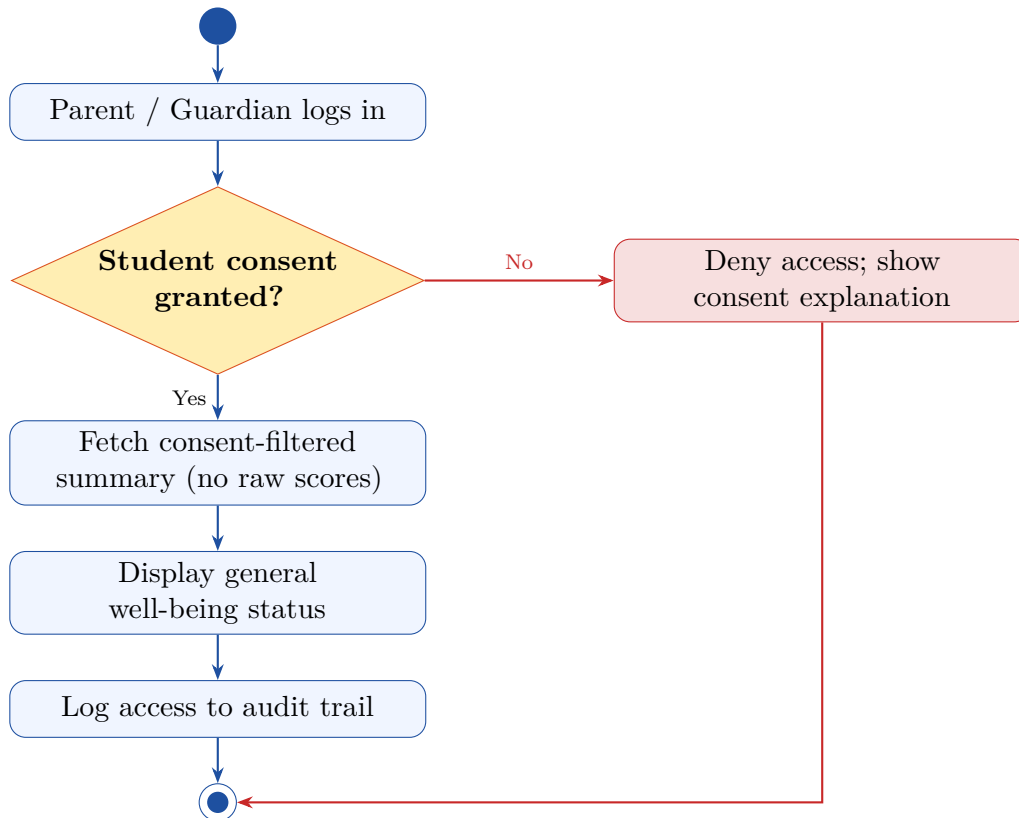


Figure 18: Activity Diagram — UC-07: View Well-being Summary

Activity Diagram — UC-08: Collect Student Data

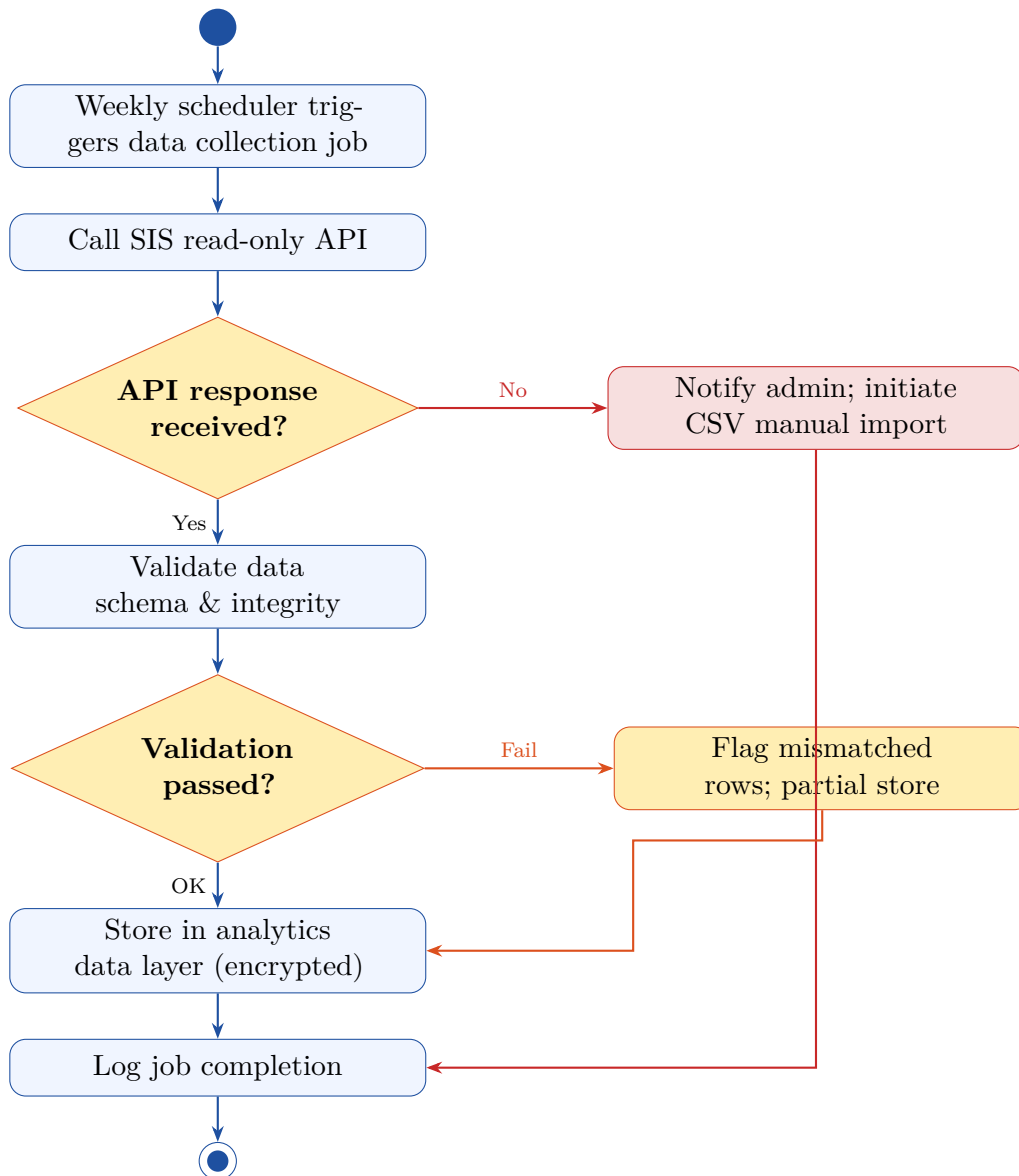


Figure 19: Activity Diagram — UC-08: Collect Student Data

3. Control & Analysis Class Diagram — UC-07 & UC-08

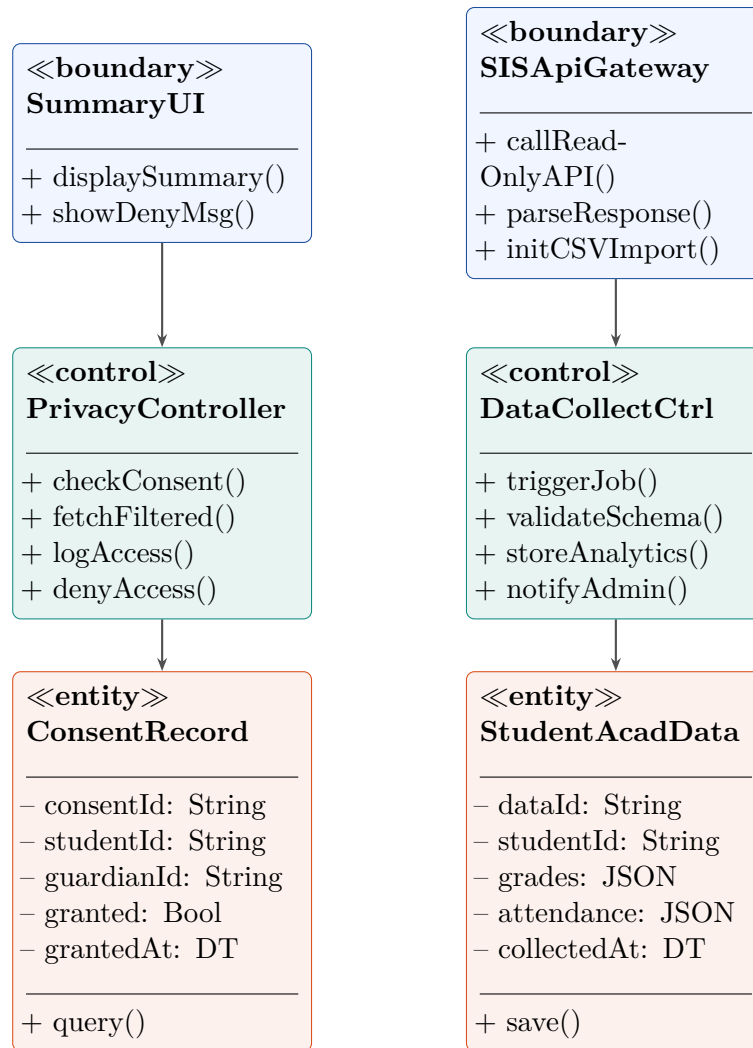


Figure 20: Analysis Class Diagram — UC-07 & UC-08

UC-09 & UC-10 — Analyse Behavioural Patterns and Detect Mental Health Risk

1. Sequence Diagrams

Sequence Diagram — UC-09: Analyse Behavioural Patterns

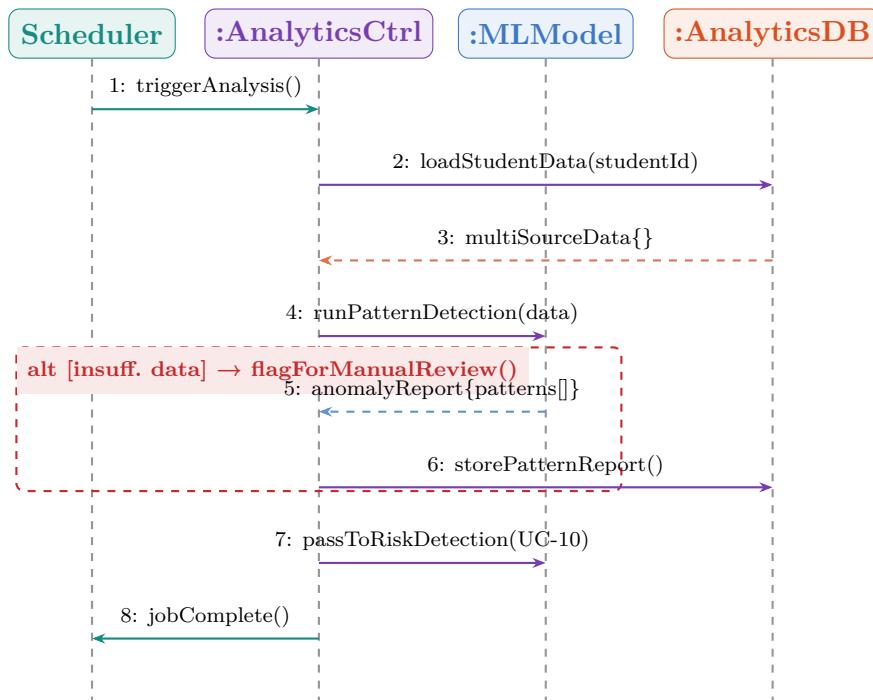


Figure 21: Sequence Diagram — UC-09: Analyse Behavioural Patterns

Sequence Diagram — UC-10: Detect Mental Health Risk (⌞include⌟ UC-09)

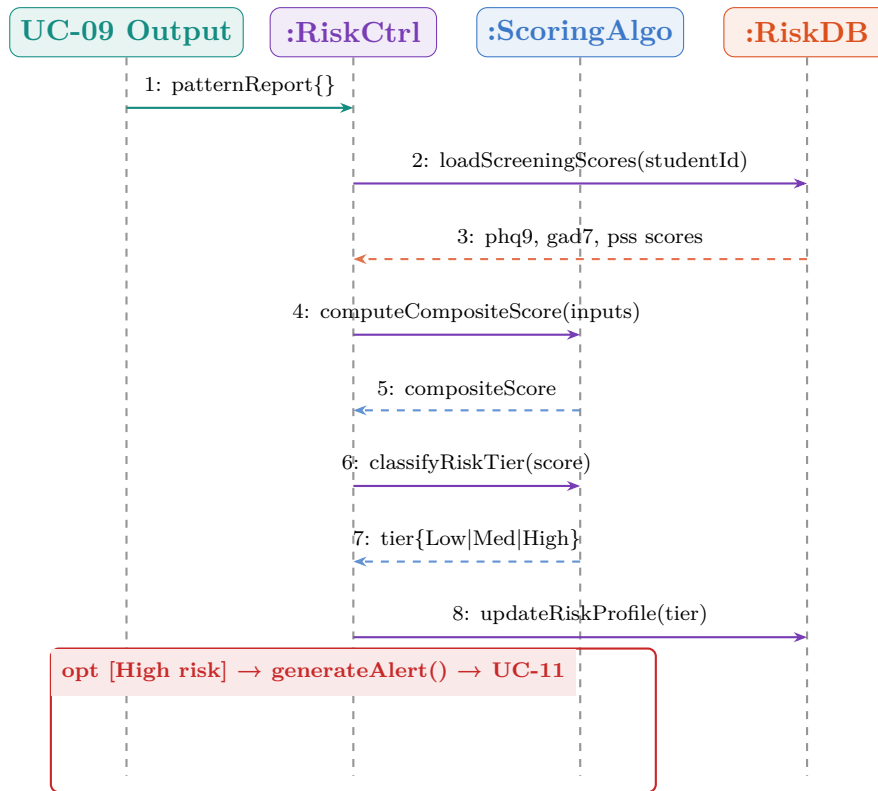


Figure 22: Sequence Diagram — UC-10: Detect Mental Health Risk

2. Activity Diagrams

Activity Diagram — UC-09: Analyse Behavioural Patterns

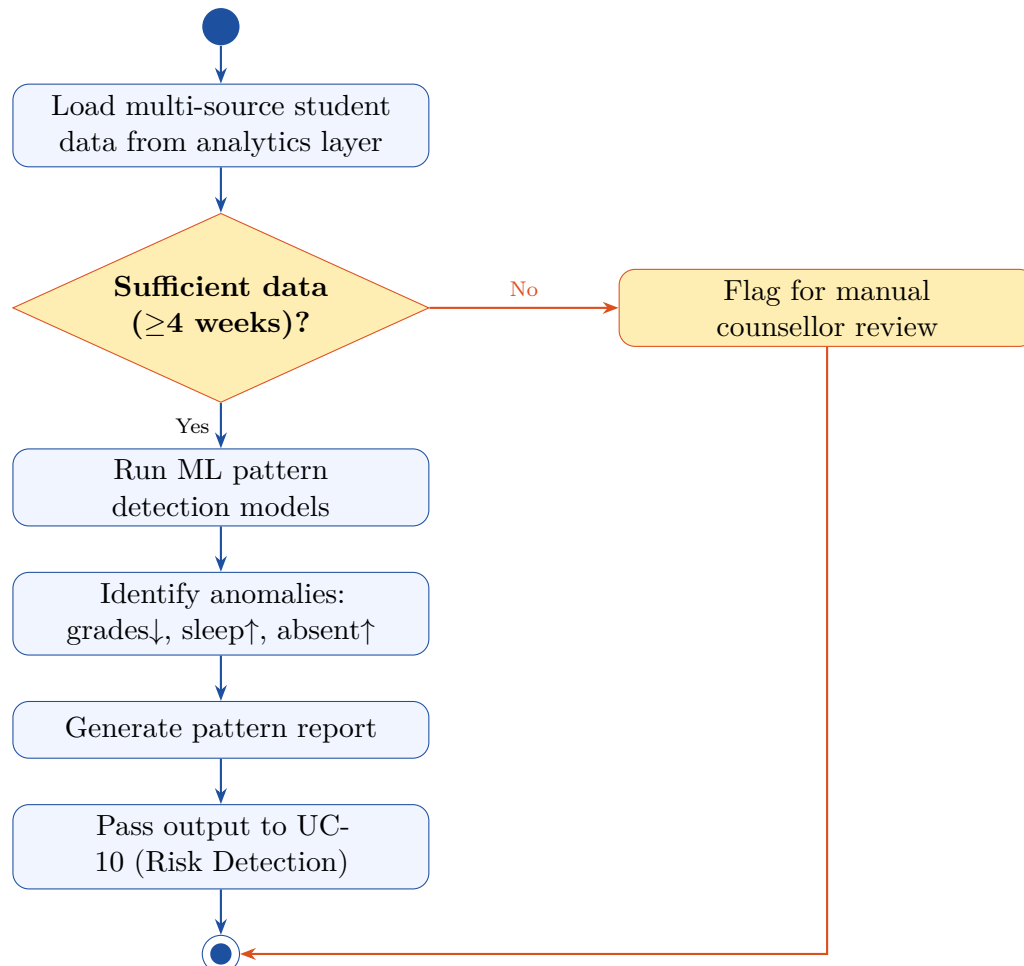


Figure 23: Activity Diagram — UC-09: Analyse Behavioural Patterns

Activity Diagram — UC-10: Detect Mental Health Risk

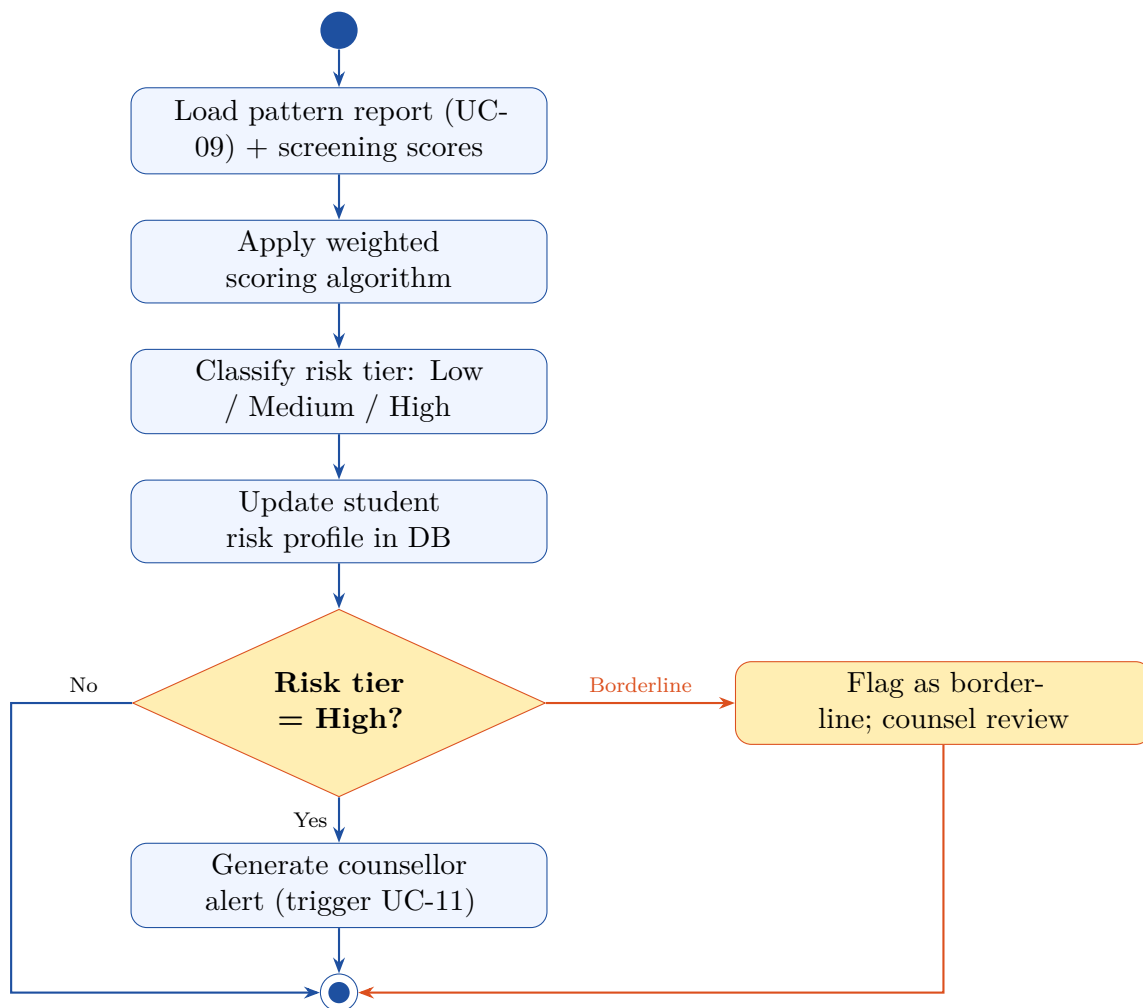


Figure 24: Activity Diagram — UC-10: Detect Mental Health Risk

3. Control & Analysis Class Diagram — UC-09 & UC-10

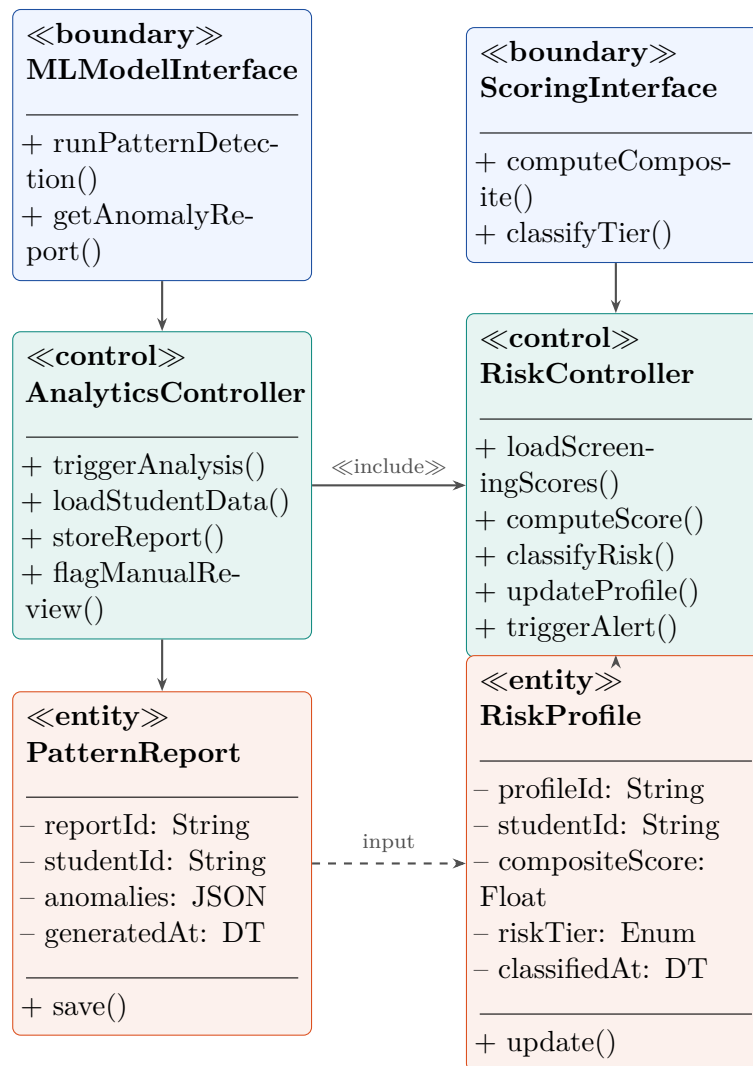
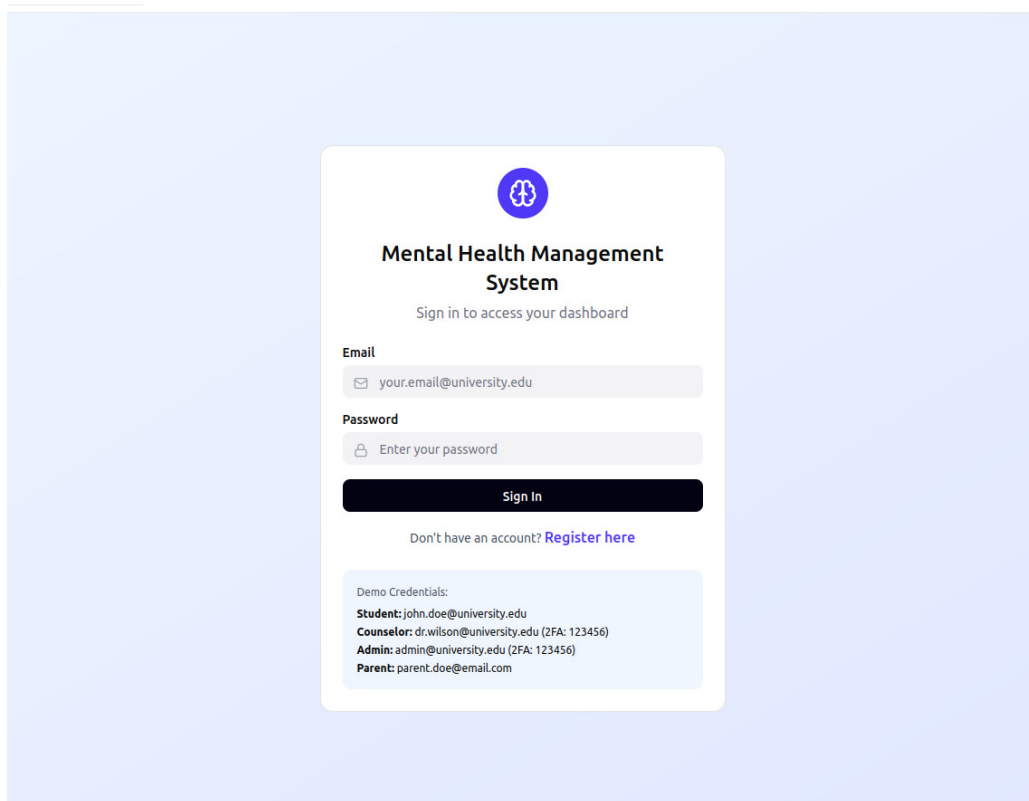


Figure 25: Analysis Class Diagram — UC-09 & UC-10

User Interface Diagrams

The following pages present the user interface screens for each actor interaction identified in the use case model. Each figure slot below corresponds to one UI screenshot; replace the placeholder frame with your actual image using `\includegraphics`.

UI-01 — Register / Login Screen



The image shows a login and registration screen for a 'Mental Health Management System'. The screen has a light blue background. In the center, there is a white card with a purple brain icon at the top. Below the icon, the title 'Mental Health Management System' is displayed in bold. Underneath the title, it says 'Sign in to access your dashboard'. There are two input fields: 'Email' with a placeholder 'your.email@university.edu' and 'Password' with a placeholder 'Enter your password'. A dark blue 'Sign In' button is below the password field. Below the button, there is a link 'Don't have an account? Register here'. At the bottom of the card, there is a section titled 'Demo Credentials:' with the following text: 'Student: john.doe@university.edu', 'Counselor: dr.wilson@university.edu (2FA: 123456)', 'Admin: admin@university.edu (2FA: 123456)', and 'Parent: parent.doe@email.com'.

Mental Health Management System

Sign in to access your dashboard

Email

your.email@university.edu

Password

Enter your password

Sign In

Don't have an account? [Register here](#)

Demo Credentials:

Student: john.doe@university.edu

Counselor: dr.wilson@university.edu (2FA: 123456)

Admin: admin@university.edu (2FA: 123456)

Parent: parent.doe@email.com

Figure 26: UI-01 — Register / Login Screen (UC-01)

UI-02 — Standardised Screening Form

← Back to Dashboard

Mental Health Screening

Complete standardized assessments to help us understand your mental health status

PHQ-9 GAD-7 PSS

Progress 0 / 9

PHQ-9 measures depression severity. Answer based on the last 2 weeks.

1. Little interest or pleasure in doing things

☐ Not at all

☐ Several days

☐ More than half the days

☐ Nearly every day

2. Feeling down, depressed, or hopeless

☐ Not at all

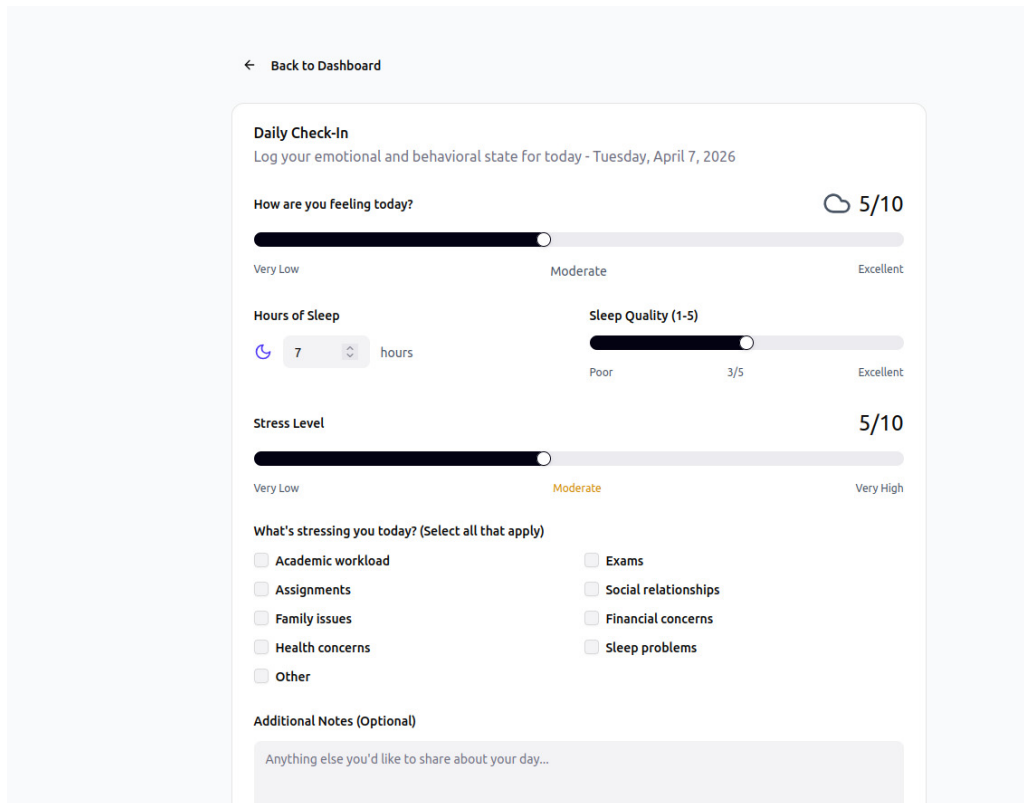
☐ Several days

☐ More than half the days

☐ Nearly every day

Figure 27: UI-02 — Standardised Screening Form (UC-02)

UI-03 — Daily Mood Check-In Screen



The image shows a mobile app interface for a 'Daily Check-In'. At the top, there is a 'Back to Dashboard' link. The main title is 'Daily Check-In' with a subtitle 'Log your emotional and behavioral state for today - Tuesday, April 7, 2026'. The first section, 'How are you feeling today?', features a horizontal slider ranging from 'Very Low' to 'Excellent', with a cloud icon and '5/10' indicating the current selection. Below this, 'Hours of Sleep' is shown as a numeric input set to '7' with a sleep icon and 'hours' label. 'Sleep Quality (1-5)' is represented by a slider from 'Poor' to 'Excellent', currently at '3/5'. The 'Stress Level' section has a slider from 'Very Low' to 'Very High', with 'Moderate' highlighted and a '5/10' score. A list of stressors follows, including 'Academic workload', 'Assignments', 'Family issues', 'Health concerns', 'Other', 'Exams', 'Social relationships', 'Financial concerns', and 'Sleep problems', each with an unchecked checkbox. At the bottom, there is an 'Additional Notes (Optional)' section with a text area labeled 'Anything else you'd like to share about your day...'.

← Back to Dashboard

Daily Check-In

Log your emotional and behavioral state for today - Tuesday, April 7, 2026

How are you feeling today? 5/10

Very Low Moderate Excellent

Hours of Sleep 7 hours

Sleep Quality (1-5) 3/5

Poor 3/5 Excellent

Stress Level 5/10

Very Low Moderate Very High

What's stressing you today? (Select all that apply)

- ☐ Academic workload
- ☐ Assignments
- ☐ Family issues
- ☐ Health concerns
- ☐ Other
- ☐ Exams
- ☐ Social relationships
- ☐ Financial concerns
- ☐ Sleep problems

Additional Notes (Optional)

Anything else you'd like to share about your day...

Figure 28: UI-03 — Daily Mood Check-In Screen (UC-03)

UI-04 — Student Mental Health Status Dashboard

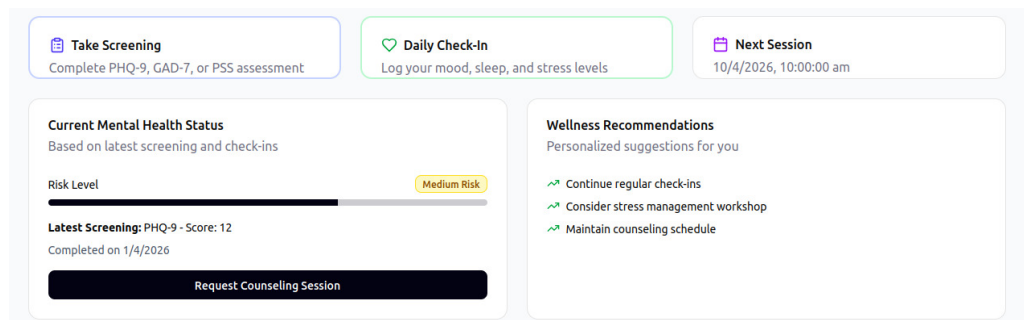


Figure 29: UI-04 — Student Mental Health Status Dashboard (UC-04)

UI-05 — Counselling Session Booking Screen

The image shows a 'Schedule Counseling Session' modal form. The form has a title bar with a close button (X). Below the title is a subtitle: 'Create a new counseling session with a student'. The form contains the following fields and controls:

- Student:** A dropdown menu with the placeholder text 'Select student'.
- Date:** A text input field with the placeholder 'dd / mm / yyyy' and a calendar icon.
- Time:** A text input field with the placeholder '--:-- --'.
- Notes (Optional):** A text area with the placeholder text 'Session notes or special considerations...'.
- Mark as urgent/high-priority case:** A checkbox.
- Buttons:** A dark blue 'Schedule Session' button and a light blue 'Cancel' button.

The background of the image shows a table with columns 'High Risk Cases' and 'Upcoming Session'. The 'High Risk Cases' column has a red '1' in the first row. The 'Upcoming Session' column has a red '2' in the first row. The table also has columns for 'Status', 'Notes', and 'Actions'. The 'Status' column has a red 'Urgent' label. The 'Notes' column has the text 'High risk alert triggered'. The 'Actions' column has a 'Complete' button. The table rows are partially visible, showing dates and times like '26, 2:00:00 pm'.

Figure 30: UI-05 — Counselling Session Booking Screen (UC-05)

UI-06 — Notification & Alert Panel

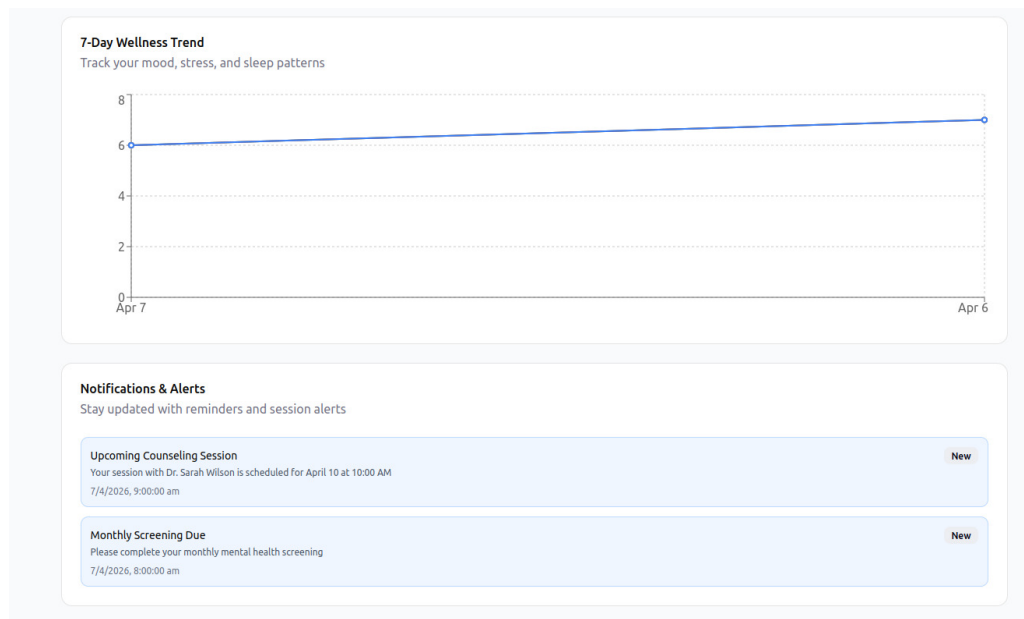


Figure 31: UI-06 — Notification & Alert Panel (UC-06)

UI-08 — System Data Collection Status Panel

The image displays a user interface for a system data collection status panel, divided into two main sections: 'At-Risk Students' and 'Counseling Sessions'.

At-Risk Students
Prioritized list with risk scores and contributing factors

Buttons: [Schedule Session](#)

Student	Risk Level	Contributing Factors	Last Assessed	Actions
Jane Smith jane.smith@university.edu	High Risk	- Academic decline - Sleep irregularity - Reduced engagement	7/4/2026	Flag Urgent
John Doe john.doe@university.edu	Medium Risk	Sleep irregularity	7/4/2026	

Counseling Sessions
Manage scheduled, urgent, and completed sessions

Student	Date & Time	Status	Notes	Actions
Jane Smith	8/4/2026, 9:00:00 am	Urgent	Flagged as urgent case requiring immediate attention	Complete Cancel
Jane Smith	8/4/2026, 9:00:00 am	Urgent	Flagged as urgent case requiring immediate attention	Complete Cancel
Jane Smith	8/4/2026, 9:00:00 am	Urgent	Flagged as urgent case requiring immediate attention	Complete Cancel
Jane Smith	8/4/2026, 2:00:00 pm	Urgent	High risk alert triggered	Complete Cancel

Notification: Student flagged as urgent - session scheduled

Figure 32: UI-08 — System Data Collection Status Panel (UC-08)

UI-10 — Risk Detection & Alert Generation Screen

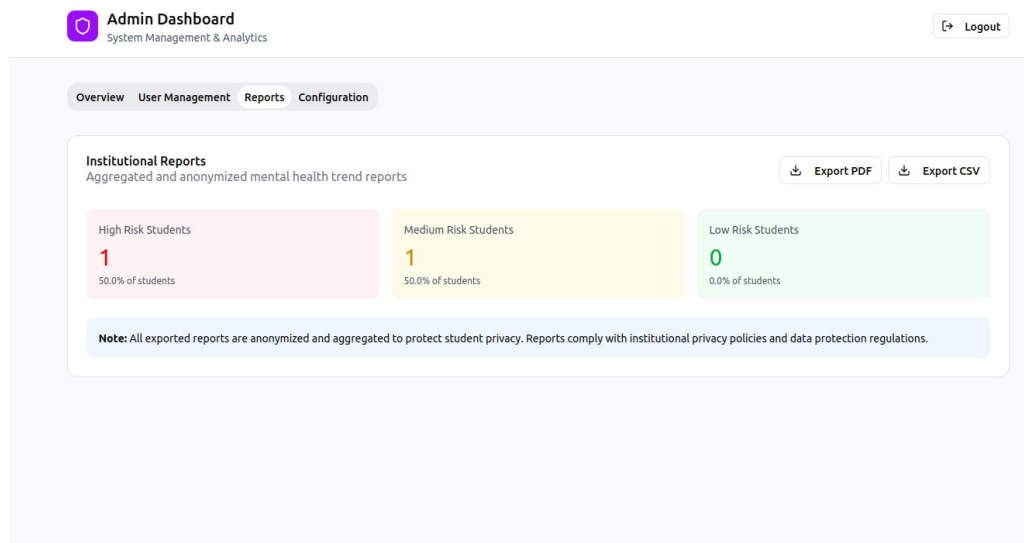


Figure 33: UI-10 — Risk Detection & Alert Generation Screen (UC-10)